NanoGT Match Report: Neurofibromatosis type 1

Disease: Neurofibromatosis type 1 (ORPHA:636)
Primary gene: NF1
Gene CDS: 8451 bp
Inheritance: Autosomal dominant
Target tissues scored: CNS, skin, peripheral nerve

Interpretation

- No packagable precedent was found in the current catalog.
- Treat this disease as out-of-scope for single-vector precedent matching until an alternative modality or engineered construct is defined.

Disease Mechanism Evidence

Molecular mechanism: haploinsufficiency
Mechanistic detail: Germline NF1 mutation reduces neurofibromin tumour suppressor dosage; subsequent somatic loss of the wild-type allele in individual cells initiates benign neurofibromas; gene addition cannot prevent independent somatic second-hit events across the lifetime of the patient
Gene-addition compatibility: conditional
Preferred modality class: gene addition tumour suppressor rescue
Evidence level/status: direct / source_checked
Evidence summary: Serra et al. (1997 Am J Hum Genet PMID 9326316) confirmed biallelic NF1 inactivation in neurofibromas by LOH analysis demonstrating the two-hit mechanism; gene addition is conditional because somatic second hits in independent cells cannot be globally prevented by a single-dose systemic vector
Evidence source: Serra et al. 1997 Am J Hum Genet PMID 9326316

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 0 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval

Excluded Programs (Packaging Failure)

Program	Vector	Gene CDS Issue
ABO-101	AAV9	Gene CDS (8451bp) exceeds vector cargo (4700bp) — hard fail
AT132	AAV8	Gene CDS (8451bp) exceeds vector cargo (4700bp) — hard fail
AVR-RD-01	LV	Gene CDS (8451bp) exceeds vector cargo (8000bp) — hard fail
BMN 307	AAV5	Gene CDS (8451bp) exceeds vector cargo (4700bp) — hard fail
CPCB-RPE1	AAV8	Gene CDS (8451bp) exceeds vector cargo (4700bp) — hard fail